Run the *mkfs.lustre* utility to create a file system named Lustre on the server including the metadata target and the management server.

Now, create a mount point and start the MDS node as shown below:

```
# mkdir /mdt
# mount -t lustre /dev/vg00/mdt /mdt
```

Run the *mount* command to verify the overall setting:

```
# mount
```

Ensure the MDS related services are enabled:

```
[root@MDS ~] # modprobe lnet
[root@MDS ~] # lctl network up
[root@MDS ~] # lctl list_nids
192.168.1.185@tcp
```

This completes MDS configuration.

Setting up the Object Storage Server (OSS1)

An Object Storage Server needs to be set up on a separate machine. I assume Lustre has been installed on a separate box and booted into the Lustre kernel. As we did earlier for MDS, we need to create several logical volumes, namely, *ost1* to *ost6*, as shown in Figure 4.

Use the *mkfs.lustre* command to create the Lustre file systems as shown below:

```
[root@oss1-0 ~] # mkfs.lustre --fsname lustre --ost
```

```
root@oss1-0 ~]#pvcreate /dev/sda2
root@oss1-0 ~]#lvcreate --name vg00/ost1 --size 6G
  Logical volume "ost1" created
[root@oss1-0 ~]# lvcreate --name vg00/ost2 --size 4G
  Logical volume "ost2" created
[root@oss1-0 ~]# lvcreate  --name vg00/ost3  --size 4G
  Logical volume "ost3" created
[root@oss1-0 ~]# lvcreate --name vg00/ost4 --size 6G
  Logical volume "ost4" created
[root@oss1-0 ~]# lvcreate --name vg00/ost5 --size 4G
  Logical volume "ost5" created
[root@oss1-0 ~]# lvcreate  --name vg00/ost6  --size 6G
  Logical volume "ost6" created
~
```

Figure 4: Creating logical volumes

```
/dev/sda1 on / type ext4 (rw)
proc on /proc type proc (rw)
sysfs on /sys type sysfs (rw)
devpts on /dev/pts type devpts (rw,gid=5,mode=620)
tmpfs on /dev/shm type tmpfs (rw)
/dev/sda1 on /boot type ext4 (rw)
none on /proc/sys/fs/binfmt_misc type binfmt_misc (rw)
/tmp on /tmp type none (rw,bind)
/var/tmp on /var/tmp type none (rw,bind)
/home on /home type none (rw,bind)
none on /proc/xenos1/config type configfs (rw)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw)
gvfs-fuse-daemon on /root/.gvfs type fuse.gvfs-fuse-daemon (rw,nosuid,nodev)
/dev/mapper/vg00-ost1 on /mnt/ost1 type lustre (rw)
/dev/mapper/vg00-ost2 on /mnt/ost2 type lustre (rw)
/dev/mapper/vg00-ost3 on /mnt/ost3 type lustre (rw)
/dev/mapper/vg00-ost5 on /mnt/ost5 type lustre (rw)
/dev/mapper/vg00-ost5 on /mnt/ost5 type lustre (rw)
/dev/mapper/vg00-ost6 on /mnt/ost6 type lustre (rw)
```

Figure 5: Output of mount

```
[root@juzreceiver1 ~]# lfs df -h
UUID                       bytes        Used   Available Use% Mounted on
lustre-MDT0000_UUID         4.8G      274.1M        3.9G   6% /mnt/lustre[MDT:0]
lustre-OST0000_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:0]
lustre-OST0001_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:1]
lustre-OST0002_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:2]
lustre-OST0003_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:3]
lustre-OST0004_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:4]
lustre-OST0005_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:5]
lustre-OST0006_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:6]
lustre-OST0007_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:7]
lustre-OST0008_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:8]
lustre-OST0009_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:9]
lustre-OST000a_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:10]
lustre-OST000b_UUID         5.9G      274.1M        5.3G   5% /mnt/lustre[OST:11]

filesystem_summary:        70.8G        3.2G       64.0G   5% /mnt/lustre
```

Figure 6: *lfs* in action

```
--mgsnode=192.168.1.185@tcp0 /dev/vg00/ost1
```

Run the above command for *ost1* to *ost6*, in a similar way. Verify the various logical volumes created, as shown below:

```
#mount -t lustre /dev/vg00/ost1 /mnt/ost1
mkfs.lustre --fsname lustre --ost --mgsnode=192.168.1.185@
tcp0 /dev/vg00/ost2
mkfs.lustre --fsname lustre --ost --mgsnode=192.168.1.185@
tcp0 /dev/vg00/ost3
mkfs.lustre --fsname lustre --ost --mgsnode=192.168.1.185@
tcp0 /dev/vg00/ost4
mkfs.lustre --fsname lustre --ost --mgsnode=192.168.1.185@
tcp0 /dev/vg00/ost5
mkfs.lustre --fsname lustre --ost --mgsnode=192.168.1.185@
tcp0 /dev/vg00/ost6
```

It's time to start the OSS by mounting the OSTs to the corresponding mount point:

```
#mount -t lustre /dev/vg00/ost2 /mnt/ost2
#mount -t lustre /dev/vg00/ost3 /mnt/ost3
#mount -t lustre /dev/vg00/ost4 /mnt/ost4
#mount -t lustre /dev/vg00/ost5 /mnt/ost5
#mount -t lustre /dev/vg00/ost6 /mnt/ost6
```

Finally, the *mount* command will display the logical volumes, as shown in Figure 5.

```
# mount
```

Verify the relative device displays as shown:

```
# cat /proc/fs/lustre/devices
```

This completes the OSS1 configuration.

Follow similar steps for OSS2 (as shown above). It is always recommended that you perform the striping over all the OSTs by running the following command on Lustre Client:

```
#lfs setstripe -c -1 /mnt/lustre
```

Setting up Lustre Client #1

All clients mount to the same file system identified by the